Fourier Series

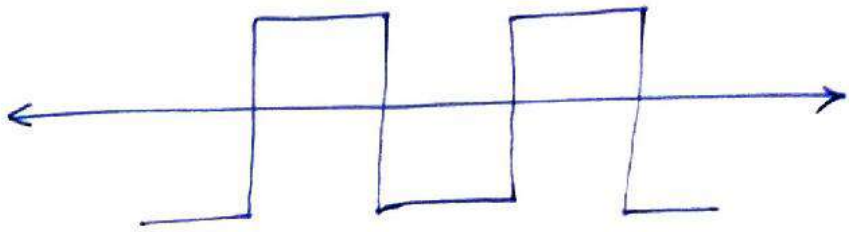
What do you expect if a cosine signal and its harmonics are added?

We must expect a periodic signal This is because all the harmonics are periodic with time period of the fundamental cosine signal.

The counter question is: can all periodic signals be written as sum of cosines and its harmonics?

Lagrange said this is propestrous: because cosine signals are continuous and sum of cosine signals signals and its harmonics cannot be discontinuous. But we can have discontinuous periodic signals.

e.g.



But for the time-being let us ignore Lagrange and assume the counter-question's answer is YES.

Underlying properties:

① Multiplying two harmonics produces a new harmonic with same fundamental frequency:

$$e^{j\omega_0 k t} \times e^{j\omega_0 m t} = e^{j\omega_0 (k+m) t}$$

② The integral of a harmonic over any time interval with length equal to a period T is zero unless the harmonic is at DC:

$$\int_{t_0}^{t_0+T} e^{jk\omega_0 t}\,dt = \int_{T} e^{jk\omega_0 t}\,dt = \begin{cases} 0, & k \neq 0 \\ T, & k = 0 \end{cases}$$

$$= T\delta[k]$$

We assume that

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 k t}$$

fundamental period

To write down sine as sum of exponential using Euler's identity.

Assume that $x(t)$ is periodic in T and is composed of a weighted sum of harmonics of $\omega_0 = 2\pi/T$.

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 k t}$$

Then

$$\int_T x(t) e^{-j\omega_0 m t} dt = \int_T \left(\sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 k t} \right) e^{-j\omega_0 m t} dt$$

$$= \sum_{k=-\infty}^{\infty} a_k \int_T e^{j\omega_0 (k-m) t} dt$$

$$= \sum_{k=-\infty}^{\infty} a_k T \delta[k-m] = T a_m$$

Therefore, $a_m = \frac{1}{T} \int_T x(t) e^{-j\omega_0 m t} dt = \frac{1}{T} \int_T x(t) e^{-j \frac{2\pi m}{T} t} d$

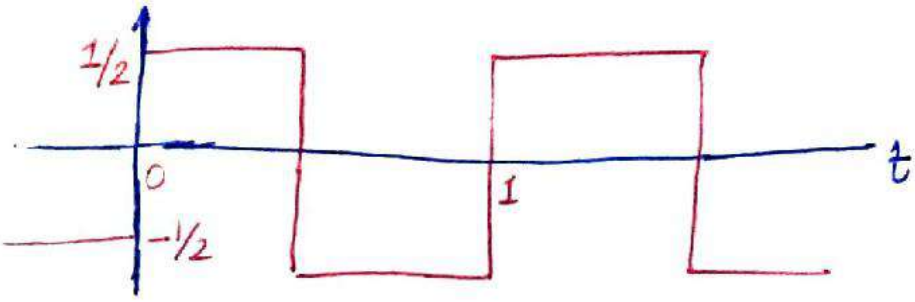
If we assume that the signal can be written as sum of harmonics, then we can find the co-efficients as.

<u>Determining harmonic components of a periodic signal</u>

$$a_k = \frac{1}{T} \int_T x(t) e^{-2\pi/T \, kt} dt \quad (\text{"analysis" equation})$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j 2\pi/T \, kt} \quad (\text{"synthesis" equation})$$

Assume this works why don't we try to find the components of harmonics for the following:



$$x(t) = x(t+1) = \sum_{k=-\infty}^{\infty} a_k e^{+j\frac{2\pi}{T}kt}$$

$$a_k = \frac{1}{T}\int_{T} x(t)\, e^{-j\frac{2\pi k}{T}t}$$

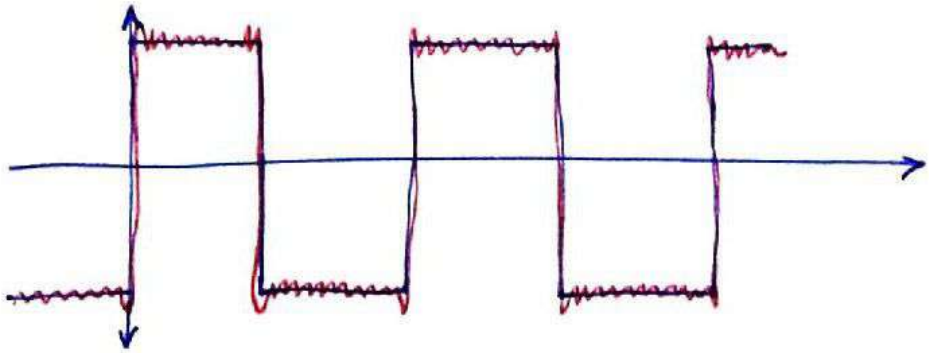
$$= \int_{-1/2}^{0} -\frac{1}{2} e^{-j2\pi k t}\, dt + \int_{0}^{1/2} 1/2\, e^{-j2\pi k t}\, dt$$

$$= \frac{-1}{j4\pi k}\left(\left[-e^{-j2\pi k t}\right]_{-1/2}^{0} + \left[e^{-j2\pi k t}\right]_{0}^{1/2} \right)$$

$$= \frac{-1}{j4\pi k}\left\{ -1 + e^{j\pi k} + e^{-j\pi k} - 1 \right\}$$

$$= \frac{1}{j4\pi k}\left\{ 2 - e^{j\pi k} - e^{-j\pi k} \right\} = \begin{cases} \frac{1}{j\pi k}, & k \text{ odd} \\ 0, & \text{even} \end{cases}$$

For the square waveform, the approximation cannot keep in accordance with the fast transitions.



Partial sums of Fourier series of discontinuous functions "ring" near discontinuities: Gibb's phenomenon.

Even in the limit the worst case deviation is 9%.

↓

Lagrange was right.

The question that remains then is when does the Fourier series converge to the original signal?

What does convergence mean in this case?